Leo J. Chung

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EDUCATION

B.Sc. in Cellular & Molecular Neuroscience **Sep. 2024 – May 2028 (Expected)**
The University of British Columbia *Vancouver, British Columbia*

TECHNICAL SKILLS

Programming	Python, R, SQL, Git, Google Apps Script, Agentic-Workflows
Analytics Tools	Microsoft Office Suite (Excel, PowerPoint), Google Workspace, GraphPad Prism, Tableau
Data Science	Statistical Modelling, Regression, Machine Learning, Web Scraping, Data Visualization
Business & Reporting	Stakeholder Presentations, Technical Reports, Product Verification, Licensing Research
Molecular Biology	RT-qPCR, PCR, RNA & DNA Extraction, cDNA Synthesis, Steroid Extraction, LC-MS, SEC, FACS
Cellular & Tissue	Cell & Organoid Culture, Primary Cell Isolation, ICC & IHC, Cryosectioning, Mouse Handling
Imaging	Confocal Microscopy, Slide Scanner Imaging, Light Microscopy, ImageJ & Fiji

CORPORATE & ACADEMIC RESEARCH EXPERIENCES

Research & Development Intern – Product Licensing, Testing, & Integrations **May 2026 – Present**
STEMCELL Technologies Inc. *Vancouver, British Columbia*

- Engineer PULSE, an AI-driven platform built in Python and Google Apps Script that scrapes 5,000+ research papers into Tableau dashboards of product usage metrics, cutting 100+ hours of manual review and informing portfolio planning.
- Build PaperTrail and PaperSift, SQL-backed AI tools version-controlled in Git that rank newly published studies by field, project relevance, and journal impact, saving scientists hundreds of hours of literature searching.
- Validate $9M+ in projected FY27 sales and support licensing decisions by planning and executing verification testing of 2 new life science products against defined acceptance criteria and troubleshooting inconsistent results.
- Maintain and passage 3 cell and organoid culture lines with aseptic technique, performing light microscopy confluency checks, media changes, and dissociation to prepare cells for downstream assays.
- Characterize organoid morphology, cell composition, and gene expression through immunocytochemistry (ICC), confocal microscopy, high-throughput RNA extraction, cDNA synthesis, and RT-qPCR.
- Purify protein and nucleic acid samples with size-exclusion chromatography (SEC) for partner research teams.
- Present PowerPoint project updates and technical reports to 20+ scientists at department-wide R&D meetings.

Research Data Analyst & Statistician **May 2026 – Present**
Johns Hopkins University, Department of Psychological & Brain Sciences (Dr. Cynthia Moss) *Baltimore, Maryland (Hybrid)*

- Analyze 10,000+ behavioural datasets with regression models in R to identify flight differences between bat subtypes.
- Apply machine-learning models to high-speed .cine video files to detect and compare behavioural patterns across trials.
- Design assays combining 3D mazes, motion-capture cameras, eye-tracking, and ultrasonic microphones to study the mechanisms behind flight trajectories, response latencies, and echolocation calls.

Shenoy Undergraduate Research Fellow in Neuroscience **Sep. 2025 – May 2026**
Djavad Mowafaghian Centre for Brain Health (Dr. Annie Ciernia) *Vancouver, British Columbia*

- Ran ultrasonic vocalization assays to collect 2,000+ mouse recordings, utilizing Python and machine-learning workflows to extract acoustic features for statistical analysis across mouse genotypes.
- Cultured primary mouse neurons and quantified axonal and dendritic arborization across 50+ images in ImageJ.
- Genotyped 100+ transgenic mice from 4 colonies via DNA extraction, PCR, and gel electrophoresis for quality control.

School of Biomedical Engineering Synergy Award Researcher **May 2025 – Aug. 2025**
Djavad Mowafaghian Centre for Brain Health (Dr. Annie Ciernia) *Vancouver, British Columbia*

- Dissected whole brains and large intestines from 30+ adult mice across 4 transgenic and microbiota-gavaged colonies.
- Cryosectioned 30+ P60 mouse brains and performed immunohistochemistry (IHC) on 300+ tissue sections to label microglia, neurons, astrocytes, and amyloid-beta plaques for slide scanner imaging.
- Quantified changes in microglial morphology and plaque pathology across transgenic Alzheimer's disease and human-IBD mouse models through computational image analysis in ImageJ and Fiji.

Undergraduate Work Learn Researcher **Jul. 2024 – Apr. 2025**

Djavad Mowafaghian Centre for Brain Health (Dr. Annie Ciernia) *Vancouver, British Columbia*

- Executed RNA extractions, cDNA synthesis, and RT-qPCR to compare hypothalamic expression of 12 immune-pathway genes between human-IBD and wild-type mouse models, running statistical tests in GraphPad Prism.
- Performed steroid extractions and assisted with LC-MS runs to measure hormone levels across mouse plasma samples.
- Isolated hypothalamic microglia and neurons by fluorescence-activated cell sorting (FACS) for cell counts and ChIP-seq.
- Managed daily lab upkeep, including inventory restocking, reagent preparation, and instrument cleaning.

LEADERSHIP & TEACHING EXPERIENCES

Medical Technology Education Researcher **Jun. 2026 – Present**

UBC School of Audiology & Speech Sciences (Dr. Stacey Skoretz) *Vancouver, British Columbia*

- Run testing trials on an interactive Virtual Reality training module that teaches clinicians dysphagia care.
- Lead quantitative analysis and data visualization in R for upcoming manuscripts comparing how speech-language pathology students rate their knowledge, skills, and confidence across 3 clinical training methods.
- Co-author peer-reviewed publications in medical education through literature reviews and manuscript revisions.

Artificial Intelligence Council Advisor **Jun. 2026 – Present**

UBC Student Artificial Intelligence Advisory Council *Vancouver, British Columbia*

- Collaborating with a 13-member institutional council advising the UBC GenAI Steering Committee on ethical, equitable, and responsible AI implementation affecting over 60,000 students.
- Consulting on the creation of campus-wide student AI surveys (N > 2,700), translating qualitative student sentiment into actionable academic policy recommendations.
- Developing the design and evaluation of university-wide digital literacy resources, contributing to the building of published institutional frameworks to best educate and provided resources to students.

Student-Directed Seminar Coordinator (ASTU 400E, Neuroaesthetics) **May 2025 – May 2026**

UBC Centre for Community Engaged Learning *Vancouver, British Columbia*

- Ideated and designed a novel course on Neuroaesthetics, which was formally offered in the second semester of the 2026 UBC Winter session for 15 upper-year undergraduate students.
- Built a 13-week syllabus with readings, lecture slides, discussion topics, and rubrics alongside Dr. Steven J. Barnes.
- Led a credited university undergraduate-level course, personally leading 20+ lectures and marking 150+ assignments across a full academic semester.

WORK EXPERIENCES

General Maintenance Handyman **Jun. 2020 – Aug. 2024**

Handyman Services *Vancouver, British Columbia*

- Completed 100+ client service orders, from installations and repairs to furniture assembly and smart appliance setup.
- Managed 8 to 10 recurring customer relationships, adapting quickly to the needs of each household.
- Quick-thinking and adaptable to respond to differing problems varying by household and client.

AWARDS & DISTINCTIONS

Undergraduate Program in Neuroscience Leadership Award (1st Place, $150 CAD)	Aug. 2026
Glenn and Annetta Agnew Science Co-op Award ($500 CAD)	Jun. 2026
Shenoy Undergraduate Research Fellowship in Neuroscience, Simons Foundation ($11,500 USD)	Aug. 2025
School of Biomedical Engineering Synergy Trainee Talk Award (1st Place, $100 CAD)	Aug. 2025
Undergraduate Program in Neuroscience Research Award (1st Place, $150 CAD)	Jul. 2025
School of Biomedical Engineering Synergy Summer Studentship ($9,500 CAD)	Apr. 2025
Faculty of Dentistry Undergraduate Summer Research Award (Declined, $9,500 CAD)	Apr. 2025
Work Learn Program Recipient ($7,500 CAD)	Sep. 2024